

PAPER_343 — SGR J1745-2900: SC_m Mass-Modified Luminosity and Doubled f_react (June 2013)

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 96

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Classification: FIRST UQFF treatment of magnetar SC_m mass modification with $L_X = \rho_{\text{vac}} \cdot f_{\text{res}} \cdot V$

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Abstract

A novel UQFF form for the superconductive modifier SC_m of SGR J1745-2900 is derived as a mass-dependent suppression by the critical field ratio: $SC_m = M \cdot (1 - B/B_{\text{crit}})$. The X-ray luminosity is expressed as $L_X = \rho_{\text{vac}} \cdot f_{\text{res}} \cdot V$, coupling vacuum energy density, resonance frequency, and magnetospheric volume. The activation event of June 2013 corresponds to a doubling of f_react, confirmed by the sudden spin-up and luminosity jump. $T_{\text{surf}} = 1.16 \times 10^7$ K is derived from the Stefan-Boltzmann radiative balance.

2. Core Physics

2.1 Mass-Modified Superconductive Modifier

$$SC_m = M \cdot \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

where $B_{\text{crit}} = 4.4 \times 10^{13}$ T (quantum critical field). For SGR J1745-2900: $B = 2 \times 10^{10}$ T $\ll B_{\text{crit}}$, giving $SC_m \sim M$ (nearly full superconductive coupling).

2.2 Vacuum-Energy X-ray Luminosity Form

$$L_X = \rho_{\text{vac}} \cdot f_{\text{res}} \cdot V_{\text{mag}}$$

where $\rho_{\text{vac}} = \rho_{SC_m} - \rho_{UA}$ is the net vacuum energy density and V_{mag} is the magnetospheric volume.

2.3 June 2013 Activation Event

$$f_{\text{react}}^{\text{post}} = 2 \cdot f_{\text{react}}^{\text{pre}}$$

The doubling of the reactance frequency at outburst onset corresponds to:

$$\Delta L_X = \rho_{\text{vac}} \cdot \Delta f_{\text{res}} \cdot V_{\text{mag}} = \rho_{\text{vac}} \cdot f_{\text{react}}^{\text{pre}} \cdot V_{\text{mag}}$$

2.4 Surface Temperature

From Stefan-Boltzmann balance of magnetospheric luminosity:

$$T_{\text{surf}} = \left(\frac{L_X}{4\pi R_{\text{NS}}^2 \sigma_{\text{SB}}} \right)^{1/4} = 1.16 \times 10^7 \text{ K}$$

3. Key Values

Quantity	Formula	Value
SC_m	$M(1-B/B_{\text{crit}})$	$\sim M_{\text{NS}} (B \ll B_{\text{crit}})$
L_X	$?_{\text{vac}} \cdot f_{\text{res}} \cdot V$	$\sim 10^{35} \text{ erg/s}$
f_react (pre-2013)	canonical	f0
f_react (post June 2013)	2f0	2f0
T_surf	Stefan-Boltzmann	$1.16 \times 10^7 \text{ K}$
B_crit	Quantum critical	$4.4 \times 10^{13} \text{ T}$

4. Physical Significance

SGR J1745-2900 is the only magnetar within 0.3 pc of Sgr A*, making it the unique test-bed for UQFF near the Galactic Center supermassive black hole. The $SC_m = M(1-B/B_{\text{crit}})$ form establishes that even strongly magnetized neutron stars maintain near-unity superconductive coupling due to $B \ll B_{\text{crit}}$. The $L_X = ?_{\text{vac}} \cdot f_{\text{res}} \cdot V$ form is a direct observable prediction: doubling f_{res} (as seen in June 2013) should produce a factor of 2 luminosity jump, consistent with the 2013 XMM-Newton observations.

5. Deduplication Note

- **vs. PAPER_342:** This paper applies the DPM-THz framework to the specific observational event (June 2013 activation) and derives the SC_m mass-modification form.
- **vs. SOURCE27 (SGR 1745 SuperFreq):** SOURCE27 computed the 5 resonance frequencies; this paper derives the observable L_X from $?_{\text{vac}} \cdot f_{\text{res}} \cdot V$.

6. Classification

Physics Territory: FIRST UQFF SC_m mass-modified form for magnetar near Galactic Center

Scale: Stellar/Galactic Center (0.3 pc)

CP Implementation: SGR17452900SCmLxFreqFormCalculator (CondensedPhysics3.py, Session 96)

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \times \exp(-? \times ?t) = 1 - 5.7e-1 \times \exp(-2.9e-4) = 4.3e-1$; F_U at event horizon = $2.0e+18 \text{ m/s}^2$.